The Avengers-Justice League Multiverse Network

Introduction

You have been assigned the task of designing a secure communication network for the unified Superheroes headquarters (AJMN). For simplicity, you have been asked to design the base networks only for 7 iconic characters. The characters are Batman, Joker, Superman, Spider-Man, Iron Man, Thor, and Captain America. The total number of agents/hosts needed for each superhero base is given below:

- Batman (Batcave) → 250 Hosts
- Joker (Arkham Hideout) → 100 Hosts
- Superman (Fortress of Solitude) → 200 Hosts
- Spider-Man (Queens HQ) → 150 Hosts
- Iron Man (Stark Tower) → 300 Hosts
- Thor (New Asgard) → 200 Hosts
- Captain America (Shield Base) → 120 Hosts

Requirements

Topology Design & Server Setup

- **Core Nodes:** Every superhero's base is represented by a router.
- **Isolated Link:** Spider-Man's base is only connected to Iron Man's base.
- **Central Hub:** Iron Man, Thor, Captain America, and Batman's bases are connected to one another via a central switch. Due to having past histories, Iron Man and Captain America have a direct connection too.
- **The Justice Loop**: There is a network loop running through the Justice Alliance bases. **Batman** is directly connected to **Superman**, **Superman** is connected to **Joker**, and then **Joker** is connected back to **Batman**.
- **Local Services:** All superhero bases have their own local Email Servers set up for secure communications. You must configure the emails accordingly. The hostnames follow the format `mail.batman.com`, `mail.joker.com`, etc.

- **Dedicated Web Servers:** Only Batman (`www.batman.justice.com`) and Iron Man (`www.ironman.avengers.com`) have dedicated web servers. You must configure the web servers so that all devices across the multiverse network can visit them.
- **Central Directory:** Only Batman's base hosts the central DNS server. You have to configure the DNS server so that all devices across all bases can use it for domain resolution.
- **Dynamic Addressing Rules:** Batman's router acts as the DHCP server for all the Justice League Bases (Batman, Joker, Superman). Ironman, Thor, Captain America bases have their own dedicated local DHCP servers. Spiderman uses Iron Man's DHCP Server because he is yet to learn, **"*With great power comes great responsibility*"**. You have to configure the servers accordingly.
- **End Devices:** Each base has a dedicated network printer, and you should display exactly two PCs to represent all the hosts/agents of that particular base.

Addressing

- Select the student ID of the first member of your group. Now take the last 4 digits of that ID and divide them into two parts containing two digits each. These two parts are the first two octets of the network address of the FIFA network. The third and fourth octets are 0, and the prefix mask is 16. Suppose the ID is 20201002. Then the last four digits are 1002. Now dividing it into two parts gives 10 and 02. So the Network address is 10.2.0.0/16.
- Subnet this network using VLSM to provide a network address to each hotel's network.
- All the PCs of the hotels will get IP addresses dynamically, while the servers, routers and printers will be configured manually.

Routing

- **Dynamic Routing:** Batman, Joker, and Superman's bases will share their routing tables with one another using the **RIP protocol**.
- **Full Awareness:** Batman's router must be aware of all possible networks in the topology, configured either dynamically or statically.

- **Default Gateway:** Spiderman can reach the rest of the multiverse networks only through Ironman's Stark Tower; therefore, a **Default Route** needs to be set up on Spiderman's router.
- **Static Mapping:** Ironman, Thor, Captain America must be configured to know about all external networks using **Static Routing**.
- **Redundancy:** If the primary link between Iron Man and Captain America goes down, Captain America must be configured to send packets through an alternate path using **Recursive Routing**.

You are allowed to make any valid and necessary assumptions while designing the network infrastructure.

Deliverables

- The network mentioned above should be implemented in Cisco Packet Tracer, with the necessary devices and full configuration.
- **In your topology, you must show the network address of each network by placing notes in Cisco Packet Tracer.**
- After completion, you should be able to test the conditions imposed, and all the devices should be able to ping one another.
- You will have to submit the following:
    - Work Distribution among the group members [Who did which part]
        - **As a member of the group, you must configure at least one of these 4 setups: DHCP/Email/Static Routing/Dynamic Routing.** Then, you can divide the rest of the work among yourselves as you see fit.
    - The pkt/pka file
    - Picture of the Network topology diagram with proper labels [You have to show the network addresses using notes for each network]
    - A PDF containing
        - VLSM tree
        - IP address table
        - The configuration commands of all the routers you have implemented.